

Lesson 4-5: Solve Rational Equations

Math Objectives	- Solve rational equations in one variable. - Identify extraneous solutions to rational equations and give examples of how they arise.
Language Objective	Students will write how to solve a rational equation word problem using a graphic organizer.
Essential Question	How can I solve rational equations and identify extraneous solutions?

CRITIQUE & EXPLAIN

Nicky and Tavon used different methods to solve the equation $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$.

Nicky	Tavon
$\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$ $\frac{1}{2}x = \frac{9}{10} - \frac{2}{5}$ $\frac{1}{2}x = \frac{5}{10}$ $x = 1$ The solution is 1.	$\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$ $10\left(\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}\right)$ $5x + 4 = 9$ $5x = 5$ $x = 1$ The solution is 1.

Explain different strategies that Nicky and Tavon used and the advantages and disadvantages of each.	
Did Nicky use a correct method to solve the equation? Did Tavon?	Use Structure. Why might Tavon have chosen to multiply both sides of the equation by 10? Could he have used another number? Explain.

HABITS OF MIND — Reason

If Tavon had multiplied each side of the equation by 100, would his answer have been 10 times as much? Explain.

What do you notice about the equation? What strategies do each of them use?

EXAMPLE 1 Solve a Rational Equation

What is the solution to each rational equation?

A. $\frac{1}{x+4} = 2$

$$\begin{aligned}(x+4)\left(\frac{1}{x+4}\right) &= 2(x+4) \\ 1 &= 2x+8 \\ x &= -\frac{7}{2}\end{aligned}$$

The solution is $x = -\frac{7}{2}$.

Multiply both sides of the equation by the common denominator to eliminate the fractions. Then solve. Confirm that the solution is valid in the original equation.

Why do you have to confirm that the solution is valid in the original equation?

Try It! Solve a Rational Equation

1. What is the solution to each equation?

a. $\frac{2}{x+5} = 4$

b. $\frac{1}{x-7} = 2$

HABITS OF MIND — Critique Reasoning

For part (b), Kaitlyn wrote $1 = 2x - 7$, then $8 = 2x$, and $4 = x$. Is she correct? Explain.

EXAMPLE 2 Solve a Work-Rate Problem

Arthur and Cheyenne are making Chinese paper cutting designs to give as gifts to wish family and friends good luck in the coming year. Together they can complete the job in 24 hours. How long would it take Cheyenne to complete the job if she was working alone? (*Cheyenne works twice as fast as Arthur.*)

Why is it important to define the variable(s) before writing an equation to represent the problem?

Try It! Solve a Work-Rate Problem

2. It takes 12 hours to fill a pool with two pipes, where the water in one pipe flows three times as fast as the other pipe. How long will it take the slower pipe to fill the pool by itself?

HABITS OF MIND — Reason

In a work-rate problem, explain why you can't average the individual rates to determine how long it will take to complete a job together.

EXAMPLE 3 Identify an Extraneous Solution

Solve the equation and identify any **extraneous solutions**:

$$\frac{1}{x-3} + \frac{x}{x+3} = \frac{2}{x^2-9}$$

Note: $x^2 - 9$ is a **difference of squares**.

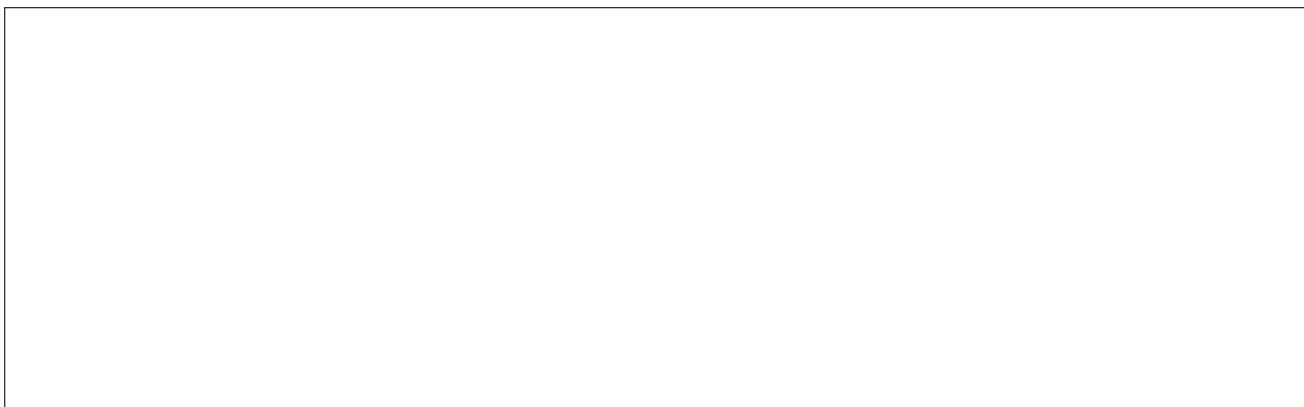
Step 1: Identify any restrictions on the domain.	
Step 2: Multiply each side of the equation by the common denominator $x^2 - 9$.	
Step 3: Solve.	
Step 4: Check for extraneous solutions.	
Step 5: Verify by graphing / extraneous solution.	

Do you have to multiply both sides of the equation by the LCD to eliminate the fractions?

How can you determine values that cannot be solutions before solving?

Try It! Identify an Extraneous Solution

3. What is the solution to the equation $\frac{1}{x+2} + \frac{1}{x-2} = \frac{4}{(x+2)(x-2)}$?



EXAMPLE 4 Check for Extraneous Solutions

What are the solution(s) to the following equation?

$$\frac{3}{x-3} = \frac{x}{x-3} - \frac{x}{4}$$

$$(4)(x-3)\left(\frac{3}{x-3}\right) = \left(\frac{x}{x-3} - \frac{x}{4}\right)(4)(x-3)$$

$$4(3) = 4(x) - x(x-3)$$

$$12 = 4x - x^2 + 3x$$

$$x^2 - 7x + 12 = 0$$

$$(x-3)(x-4) = 0$$

$$x-3=0 \text{ or } x-4=0$$

$$x=3 \text{ or } x=4$$

Multiply by the LCD.

Write in standard form so the quadratic is equal to 0. Then factor the polynomial and solve for x .

Check the solutions in the original equation. The value 3 is an **extraneous solution** because it would cause the denominator of the original equation to be equal to zero. The only solution to the equation is $x = 4$.

How could you use a graph to verify your solution?

Try It! Check for Extraneous Solutions

4. What are the solutions to the following equations?

a. $x + \frac{6}{x-3} = \frac{2x}{x-3}$

b. $\frac{x^2}{x+5} = \frac{25}{x+5}$

EXAMPLE 5 Solve a Rate Problem

Paddling with the current in a river, Jake traveled 16 miles. Even though he paddled upstream for an hour longer than the amount of time he paddled downstream, Jake could only travel 6 miles against the current. In still water, Jake paddles at a rate of 5 mph. What is the speed of the current in the river?

Formulate	
Compute	
Interpret	

Try It! Solve a Rate Problem

5. Three people are planting tomatoes in a community garden. Marta takes 50 minutes to plant the garden alone, Benito takes x minutes, and Tyler takes $x + 15$ minutes. If the three of them take 20 minutes to finish the garden, how long would it have taken Tyler alone?

HABITS OF MIND — Make Sense and Persevere

What does the fraction $\frac{1}{50}$ mean with regards to Marta?

CONCEPT SUMMARY Solving Rational Equations

WORDS

A rational equation is an equation that contains a rational expression. To solve, identify the domain for the variable. Then multiply both sides of the equation by a common denominator and solve. An **extraneous solution** is a solution that is not valid because that value is excluded from the domain of the original equation.

ALGEBRA

Domain: $x \neq 0$

$$\begin{aligned}\frac{1}{x} + \frac{2}{x} &= \frac{1}{6} \\ 6x\left(\frac{1}{x} + \frac{2}{x}\right) &= 6x\left(\frac{1}{6}\right) \\ 6 + 12 &= x \\ 18 &= x\end{aligned}$$

The domain includes $x = 18$, so the solution to the equation is 18.

Domain: $x \neq 1$

$$\begin{aligned}\frac{x^2 + 4}{x - 1} &= \frac{5}{x - 1} \\ (x - 1)\left(\frac{x^2 + 4}{x - 1}\right) &= (x - 1)\left(\frac{5}{x - 1}\right) \\ x^2 + 4 &= 5 \\ x^2 &= 1 \\ x &= \pm 1\end{aligned}$$

The domain does not include $x = 1$, so 1 is an extraneous solution. It does include $x = -1$, so the solution to the equation is -1 .

Select Exercises

8. Reason. If you solve a work-rate problem and your solution, which represents the amount of time it would take <i>working together</i>, exceeds the individual <i>working alone</i> times that are given, then how do you know your solution is unreasonable? Explain. DOK3	
20. Russel and Aaron can build a shed in the time shown when working together. Aaron works three times as fast as Russel. How long would it take Russel to build the shed if he were to work alone? (<i>Russel and Aaron can build the shed in 8 hours working together.</i>) DOK2	
Solve the equation. (SEE EXAMPLE 3) 21. $\frac{x}{x-3} - 4 = \frac{3}{x-3}$ DOK2	
Solve the equation. (SEE EXAMPLE 4) 23. $\frac{4}{3(x+1)} = \frac{12}{x^2-1}$ DOK2	

30. Generalize. So far this baseball season, Philip has gotten a hit 8 times out of 40 at-bats. He wants to increase his batting average to 0.333. Calculate the number of consecutive hits, h , he would need in order to achieve this goal. Round your answer to the nearest whole number.

DOK3